Ndemy Capstone: Resilient Raft Key-Value Store

Specification, Test Harness, Network Partition Matrix & Grading Rubric

Published by Ndemy Distributed Systems Academy | Staff Specialization

1. Objective & Requirements

Implement Raft Consensus State Machine in Go or Rust

Support dynamic cluster membership changes [Joint Consensus]

Pass the Jepsen fault-injection linearizability suite

2. Fault Scenarios Tested

Split-Brain isolation: 2 nodes separated from 3 nodes

Cascading leader crash during uncommitted log replication

Asymmetric network delays and packet drop simulation